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TIC 402710280.01: A Sub-Neptune Candidate on a 36.34-Day Orbit around a Nearby K Dwarf, Identified in TESS Full-Frame Images

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ABSTRACT

We report a transiting planet candidate around TIC 402710280 (TYC 9041-03677-1), a $T = 9.24$ K dwarf at 66.8 pc that the Transiting Exoplanet Survey Satellite (*TESS*) has observed only in full-frame images. Four transits, recorded in Sectors 39, 93, 100 and 102 and reduced by three independent pipelines, share a depth of 2097 ± 46 ppm and a duration of 5.27 ± 0.05 h. Two of the transits fall on consecutive orbits, and the predicted transits of every shorter period are absent from the data at more than 18σ; the orbital period is $P = 36.33990 \pm 0.00053$ d. A transit model with each event centred on its measured mid-time yields $R_p/R_\star = 0.0421^{+0.0006}_{-0.0005}$, impact parameter $b = 0.22^{+0.19}_{-0.15}$ and a stellar density of $1.57^{+0.11}_{-0.29}\,\rho_\odot$, in agreement with the catalogue value of $1.61 \pm 0.38\,\rho_\odot$. With $R_\star = 0.817 \pm 0.047\,R_\odot$ the candidate has a radius of $3.75 \pm 0.22\,R_\oplus$, receives 10.2 ± 1.7 times Earth's insolation and has an equilibrium temperature of 498 ± 21 K. Difference-image centroids locate the signal within $1.5'' \pm 1.3''$ of the target; the only catalogued star that could produce the signal, a background source $3.5''$ away, is disfavoured by the transit shape at $\Delta\chi^2 \geq 142$. No secondary eclipse is detected (-54 ± 39 ppm), and `TRICERATOPS` returns a false-positive probability of 0.003 ± 0.004. The mid-times scatter about a linear ephemeris by 21.5 min (rms), which we report without interpretation. We publish predicted transit times through 2028, including a transit expected in *TESS* Sector 114 on 2027 February 19.

Keywords: Exoplanets (498) — Transit photometry (1709) — Mini Neptunes (1063) — K dwarf stars (876)

1. INTRODUCTION

Transiting planets with periods longer than a month around bright, nearby stars remain scarce, because a single *TESS* sector of about 27 d (Ricker et al. 2015) records at most one of their transits. Such planets are found by combining sectors observed years apart, a task that becomes tractable as the mission's full-frame images accumulate. Stars observed only in full-frame images are additionally excluded from the Science Processing Operations Center (SPOC) search of two-minute data (Jenkins et al. 2016), so their long-period transiters depend on full-frame light curves such as those of the Quick-Look Pipeline (QLP; Huang et al. 2020a,b) and TESS-SPOC (Caldwell et al. 2020a), and on photometry of the most recent images, which these products do not yet cover.

EMAIL

Here we report TIC 402710280.01, a sub-Neptune candidate with a 36.34-day period around a K dwarf at 67 pc. Its first two transits were identified in public full-frame light curves; the period was determined from our own photometry of the 2026 full-frame images. As of 2026 October 14, the star is neither a *TESS* Object of Interest (TOI; Guerrero et al. 2021) nor a community TOI, and no planet parameters, follow-up observations or literature references are associated with it in the Exoplanet Follow-up Observing Program (ExoFOP) archive, the NASA Exoplanet Archive (Akeson et al. 2013), or the NASA Astrophysics Data System.

2. HOST STAR

Table 1 lists the properties of TIC 402710280. We adopt the stellar parameters of the *TESS* Input Catalog v8.2 (TIC; Stassun et al. 2019). *Gaia* DR3 (Gaia Collaboration et al. 2016, 2023) reports a parallax of 14.964 ± 0.014 mas, a renormalised unit weight error (RUWE; Lindegren et al. 2021) of 1.00, no multi-peaked image windows (`ipd_frac_multi_peak` = 0), and no

Table 1. Properties of TIC 402710280

Parameter	Value	Source
R.A. (J2000)	16:34:05.67	TIC
Decl. (J2000)	−63:32:58.8	TIC
Gaia DR3 ID	5828441289812582400	*Gaia*
$\mu_\alpha \cos\delta$ (mas yr^{-1})	59.909	*Gaia*
μ_δ (mas yr^{-1})	13.685	*Gaia*
Parallax (mas)	14.964 ± 0.014	*Gaia*
Distance (pc)	66.83 ± 0.06	*Gaia*
RUWE	1.00	*Gaia*
T (mag)	9.243 ± 0.006	TIC
G (mag)	9.781	*Gaia*
V (mag)	9.982 ± 0.004	TIC
K_s (mag)	8.026	TIC
$T_{\rm eff}$ (K)	5182 ± 121	TIC
$\log g$ (cgs)	4.56 ± 0.08	TIC
$R_\star$ ($R_\odot$)	0.817 ± 0.047	TIC
$M_\star$ ($M_\odot$)	0.88 ± 0.11	TIC
$\rho_\star$ ($\rho_\odot$)	1.61 ± 0.38	TIC

non-single-star solution. Together these indicate a single, well-behaved astrometric source, although they do not exclude a companion at small separation and low mass ratio.

3. OBSERVATIONS AND PHOTOMETRY

TESS observed TIC 402710280 in Sectors 12 (2019), 39 (2021), 66 (2023), 93 (2025) and 100–103 (2026), always in full-frame images, at cadences of 1800 s (prime mission), 600 s (first extended mission) and 200 s thereafter. We use the TESS-SPOC light curves (Caldwell et al. 2020a,b) for Sectors 12, 39 and 66 (pre-search data conditioning flux) and the QLP light curve (Huang et al. 2020a,c) for Sector 93 (systematics-corrected flux). Neither pipeline had processed Sectors 100–103 when this work was done, so we extracted their light curves from the calibrated full-frame images: 21 × 21-pixel cutouts (Brasseur et al. 2019), a per-cadence background equal to the median of the faintest 40% of pixels away from the target, a circular aperture whose radius (between 1 and 4 pixels) minimises the one-hour scatter of the detrended light curve, and rejection of cadences flagged by SPOC or affected by scattered-light excursions of the background.

For transit analysis, each event is extracted in a window of ±0.8 d, cadences with non-zero quality flags are removed, and the flux is normalised by a quadratic function of time fitted to the out-of-transit data. For timing, the baseline is instead refitted simultaneously with the transit model (§5.1).

4. DETECTION AND PERIOD

A search of public full-frame light curves for individual transit-like events (box filters at ten trial durations, with thresholds set by the measured scatter of each filtered series) identified two dips of matching depth and duration in Sectors 39 and 93, separated by 1453.58 d. Our photometry of Sectors 99–108 then revealed two further transits, in Sectors 100 and 102, separated by a single orbit of 36.36 d (Figure 1). Because these two transits are consecutive, the period cannot be a multiple of 36.36 d; the remaining possibilities are its integer fractions. We tested $P/2$ through $P/5$ by measuring the flux at every transit time they predict within the data, excluding the four observed events: the shallowest exclusion among the 41 covered epochs is 18.6σ (for $P/4$), and every alias is rejected. Sector 101 contains no transit at the adopted period, Sector 103's predicted transit falls in a data gap, and Sector 12 contains no predicted transit. The four transits thus fix the period uniquely at 36.34 d.

The transit predicted in Sector 66 (BTJD 3111.40, where BTJD = $\mathrm{BJD_{TDB}} - 2457000$) coincides with an interval of strong scattered light (Figure 2). The background falls from 1.5×10^4 to 2.9×10^3 e$^-$ s^{-1} across the window, quality-flagged cadences end only 3.2 h before the predicted mid-time (0.6 h before ingress), and QLP flags every cadence within 12 h of it. A transit model fitted at the predicted time returns a depth of 1620 ± 200 ppm, consistent with the other events, but the result depends on the treatment of the one-sided baseline. We therefore exclude Sector 66 from all fits; it neither supports nor contradicts the candidate.

5. TRANSIT ANALYSIS

5.1. *Individual Transit Times and Depths*

Each transit is timed by fixing the shape to the median model of §5.2 and scanning the mid-time on a 0.5-minute grid over ±2 h, refitting a quadratic baseline at each step. The 1σ uncertainty is the half-width of the $\Delta\chi^2 = 1$ interval, multiplied by the red-noise factor β (Pont et al. 2006; Winn et al. 2008), the median ratio of the binned residual scatter (10–60-minute bins) to its white-noise expectation; β ranges from 1.00 to 1.53. The depth of each event is measured with the shape fixed and a free amplitude. Table 2 lists the results.

The four depths agree ($\chi^2 = 2.3$ for 3 degrees of freedom) despite coming from three independent reductions with different apertures and cadences, which argues

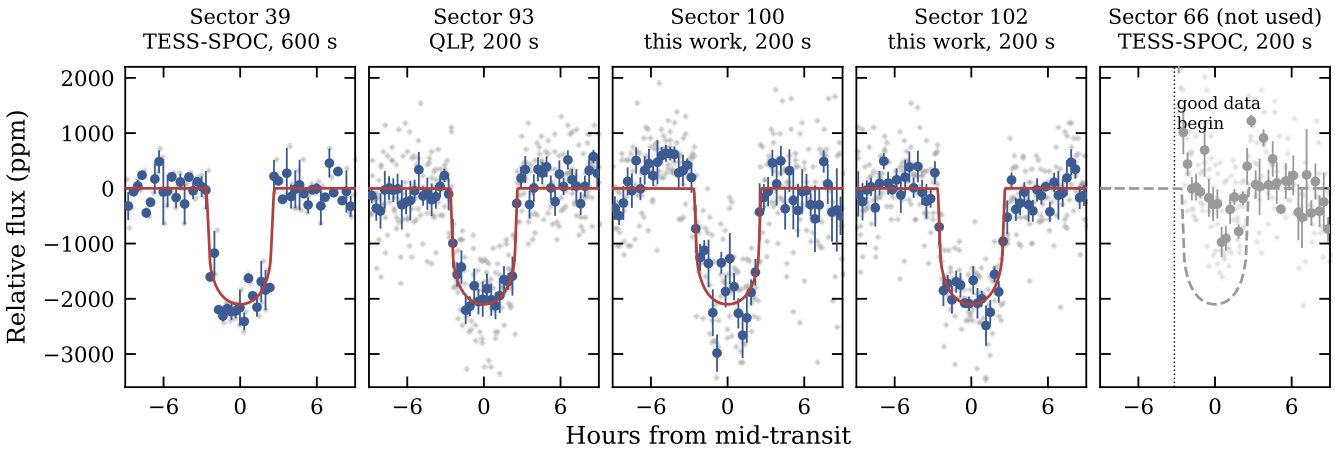


Figure 1. Transits of TIC 402710280.01. Grey points are individual cadences; blue points are 20-minute bins with their standard errors. Each window is centred on its measured mid-time (Table 2), and the red curve is the median transit model of §5.2. The right-hand panel shows the Sector 66 window at the time predicted by the linear ephemeris; data with zero quality flags begin at the dotted line, and the window is not used in any fit (§4).

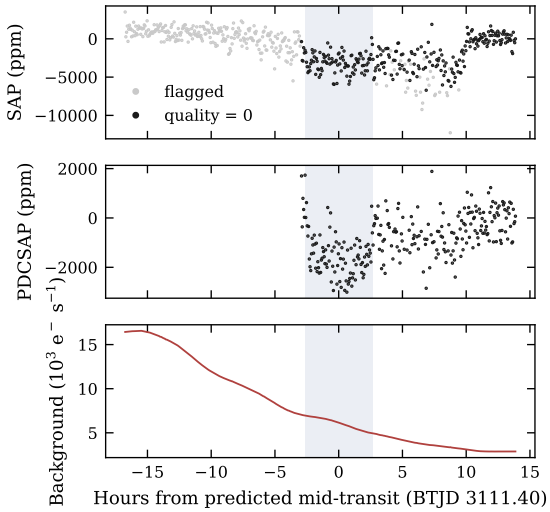


Figure 2. TESS-SPOC data of Sector 66 around the predicted transit (shaded). From top: simple-aperture flux, with quality-flagged cadences in grey; pre-search data conditioning flux, quality 0 only; and the background. The window lies on a scattered-light ramp and is not used in the analysis.

against an origin in any one pipeline. Their weighted mean is 2097 ± 46 ppm.

5.2. *Transit Shape*

We model the four transits jointly with `batman` (Kreidberg 2015), integrating over each light curve's exposure time and centring each window on its measured mid-time, so that timing scatter (§5.3) does not smear the stacked profile. The free parameters are $R_p/R_\star$, b and $\log_{10}\rho_\star$, with uniform priors; $a/R_\star$ follows from $\rho_\star$ and P (Seager & Mallén-Ornelas 2003) for a circular orbit. Quadratic limb-darkening coefficients are fixed at $(u_1, u_2) = (0.39, 0.21)$, interpolated in $T_{\rm eff}$ from the *TESS*-band tables of Claret (2017) at $\log g = 4.5$. Each window's flux uncertainty is its out-of-transit point-to-point scatter multiplied by its β. We sample the posterior with `emcee` (Goodman & Weare 2010; Foreman-Mackey et al. 2013), using 40 walkers for 30 000 steps and discarding the first third; the chain spans more than 100 integrated autocorrelation times for every parameter. Table 3 gives the medians and 68% intervals.

The fitted stellar density, $1.57^{+0.11}_{-0.29}\,\rho_\odot$, agrees with the TIC value of $1.61 \pm 0.38\,\rho_\odot$ within 0.1σ; the transit is therefore consistent with a planet on a circular orbit around TIC 402710280 itself. The best-fitting model has $\chi^2 = 1866$ for 2227 data points.

5.3. *Ephemeris and Timing Residuals*

A weighted linear fit to the four mid-times gives $P = 36.33990$ d and a reference epoch $T_0 = 2460838.2019$ BJD$_{\rm TDB}$, placed at the Sector 93 transit where the covariance between T_0 and P is smallest. The residuals (Table 2, Figure 3) are +4.8, −24.8, +5.7 and +34.3 min, with an rms of 21.5 min, much larger than the 1.7–3.5-min formal uncertainties ($\chi^2 = 461$ for 2 degrees of freedom). We scale the ephemeris uncertainties by $\sqrt{\chi^2/\nu}$, which yields $\sigma_P = 0.00053$ d and $\sigma_{T_0} = 0.0122$ d.

The scatter has two possible origins: timing systematics that the formal uncertainties do not capture, or transit-timing variations induced by an additional body. The largest residuals belong to Sector 93, a public QLP light curve, and Sector 102, whose transit occurs 1.4 d after the start of an orbit, where a residual ramp is vis-

Table 2. Individual transits of TIC 402710280.01

Sector	Light curve	Cadence (s)	Epoch	T_c ($\mathrm{BJD_{TDB}}$)	$O-C$ (min)	Depth (ppm)
39	TESS-SPOC	600	0	2459384.6094 ± 0.0013	+4.8	2203 ± 85
93	QLP	200	40	2460838.1847 ± 0.0012	−24.8	2050 ± 82
100	This work	200	47	2461092.5851 ± 0.0024	+5.7	2075 ± 150
102	This work	200	48	2461128.9449 ± 0.0016	+34.3	2045 ± 86

NOTE—$O-C$ is relative to the linear ephemeris of Table 3. Depths are the model depth at mid-transit for a free amplitude with the shape fixed.

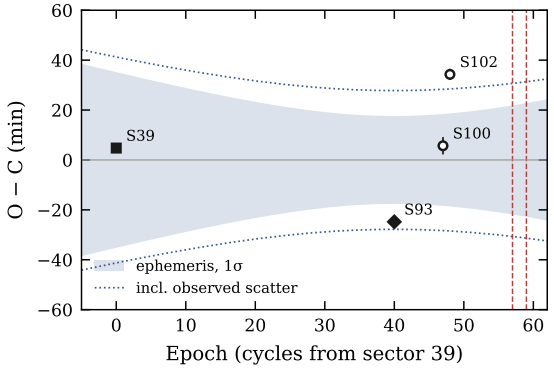


Figure 3. Transit-timing residuals relative to the linear ephemeris. Symbols denote the light-curve source: TESS-SPOC (square), QLP (diamond) and our photometry (open circles). The shaded band is the 1σ uncertainty of the ephemeris; the dotted curves add the 21.5-min rms scatter of the residuals in quadrature. Dashed lines mark epochs 57 (2027 February 19, *TESS* Sector 114) and 59 (2027 May 3).

ible; no single reduction is therefore responsible. Four epochs cannot distinguish the two explanations, and we draw no conclusion about transit-timing variations. The predictions in §8 include the observed scatter.

Table 3. Parameters of TIC 402710280.01

Parameter	Value
Ephemeris	
P (d)	36.33990 ± 0.00053
T_0 ($\mathrm{BJD_{TDB}}$)	2460838.2019 ± 0.0122
Timing residual rms (min)	21.5
Fitted	
$R_p/R_\star$	$0.04208^{+0.00064}_{-0.00051}$
b	$0.22^{+0.19}_{-0.15}$
$\rho_\star$ ($\rho_\odot$)	$1.57^{+0.11}_{-0.29}$
Derived	
Depth at mid-transit (ppm)	2097 ± 46
T_{14} (h)	$5.27^{+0.05}_{-0.04}$
$a/R_\star$	$53.7^{+1.2}_{-3.5}$
R_p ($R_\oplus$)	3.75 ± 0.22
a (AU)	0.206 ± 0.008
S ($S_\oplus$)	10.2 ± 1.7
$T_{\rm eq}$ (K)	498 ± 21

NOTE—$T_{\rm eq}$ assumes zero Bond albedo and full heat redistribution. R_p, a, S and $T_{\rm eq}$ use the stellar parameters of Table 1.

6. VETTING

6.1. *Location of the Signal*

For each transit we form a difference image (out-of-transit minus in-transit) from the full-frame image cutouts and fit a Gaussian point-spread function to it (cf. Bryson et al. 2013; Twicken et al. 2018). Each offset is measured relative to the target's own fitted position in the out-of-transit image, which removes errors in the image's world-coordinate solution and the effect of proper motion. Combining the four transits gives an offset of $1.5'' \pm 1.3''$ from TIC 402710280, consistent with the target at 1.2σ. We validated the method on 40 confirmed *TESS* planets: it placed 30 on their host star, left 5 unresolved from a nearby source, returned an offset without a catalogued counterpart for 3, failed for 2, and placed none on a neighbouring star.

Every *Gaia* DR3 source bright enough to produce the observed depth, even if fully eclipsed, lies 21σ or more from the measured position, with one exception: *Gaia* DR3 5828441289812582656 ($G = 16.15$, $\Delta G = 6.37$), whose parallax of 0.69 ± 0.08 mas places it in the background. Owing to the target's proper motion, its separation increases from $3.10''$ at the *Gaia* epoch (2016.0) to $3.4''$–$3.6''$ during the transits; the centroid lies 1.6σ from its position at the mean transit epoch, so the difference images alone cannot exclude it.

6.2. *Blended Eclipsing Binary*

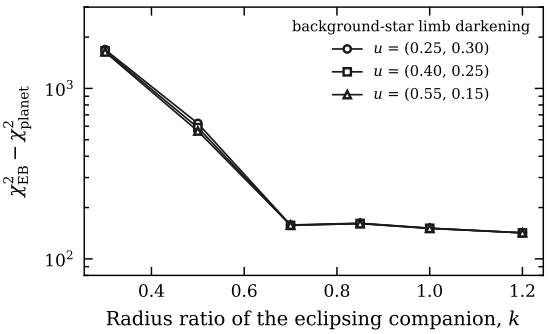


Figure 4. Diluted-eclipse models for the background star *Gaia* DR3 5828441289812582656, compared with the planet model. Each point is the minimum χ^2 difference over impact parameter, $a/R_\star$ and dilution for a fixed companion radius ratio k; symbols denote the assumed limb darkening of the background star.

Diluted by TIC 402710280, this star could reproduce the observed depth only through an eclipse of about 74% of its own light, which requires a companion of comparable size and produces long ingress and egress. We fitted the four transits with eclipse models of the background star for companion-to-star radius ratios $k = 0.3$ to 1.2, with the impact parameter and $a/R_\star$ free, the dilution free but bounded at 0.4% of the total flux (40% above the star's share, to allow for its unknown *TESS*-band colour), and three limb-darkening laws. Every such model is worse than the planet model by $\Delta\chi^2 \geq 142$ for 2227 data points (Figure 4). A blended eclipsing binary on this star is therefore strongly disfavoured, although high-resolution imaging is required to exclude unresolved sources closer than about 1″.

6.3. *Secondary Eclipse and Odd–Even Depths*

The light curve at orbital phase 0.5, sampled in Sectors 39, 101 and 102, shows no eclipse: the weighted depth is -54 ± 39 ppm, below 62 ppm at 3σ, which excludes an eclipsing companion of comparable surface brightness on a circular orbit. A comparison of odd and even transits is not informative, because only one of the four transits (Sector 100) falls on an odd epoch; the agreement of all four depths (§5.1) nonetheless disfavours an eclipsing binary at twice the period with unequal eclipses.

6.4. *Statistical False-Positive Probability*

We computed scenario probabilities with `TRICERATOPS` (Giacalone & Dressing 2020; Giacalone et al. 2021), using the four transits stacked on their measured mid-times (10-minute bins), a depth of 2100 ppm, the photometric aperture of each sector, and the *Gaia* DR3 field. Ten independent runs give a false-positive probability FPP $= 0.003 \pm 0.004$ (largest value 0.012) and a nearby false-positive probability NFPP $= 4 \times 10^{-24}$. The planet hypothesis on the target carries 85% of the probability and a planet diluted by an unresolved bound companion 14%; the leading false-positive scenario, a planet orbiting an unresolved companion, carries 0.3%. With the larger default 5×5-pixel apertures, FPP $= 0.001$ and NFPP $= 6 \times 10^{-7}$. These values satisfy the numerical criteria of Giacalone et al. (2021), FPP < 0.015 and NFPP < 0.001, but statistical validation also requires high-resolution imaging, which does not yet exist for this star. We therefore classify TIC 402710280.01 as a planet candidate.

7. DISCUSSION

With $R_p = 3.75 \pm 0.22\,R_\oplus$, TIC 402710280.01 lies above the radius valley (Fulton et al. 2017), among sub-Neptunes that likely retain a substantial volatile envelope. At 0.206 AU it receives ten times Earth's insolation. The mass–radius relation of Chen & Kipping (2017) predicts $M_p \approx 14\,M_\oplus$ and a radial-velocity semi-amplitude of $K \approx 2.8$ m s^{-1}, an amplitude accessible to current high-precision spectrographs for a $V = 10.0$ star.

Three observations would most efficiently confirm the planet: high-resolution imaging, to exclude unresolved companions and complete the statistical validation; additional transits, to refine the ephemeris and test the timing residuals; and radial velocities, to measure the mass. The transit-timing scatter (§5.3) is itself a target for follow-up, since a nearby perturber would make the system a candidate for dynamical mass measurement.

8. PREDICTED TRANSITS

Table 4 lists the next transits predicted by the ephemeris of Table 3. The quoted uncertainty, $\sigma_{\rm pred} = (\sigma_{T_0}^2 + n^2\sigma_P^2 + \sigma_{\rm rms}^2)^{1/2}$, adds the observed residual scatter $\sigma_{\rm rms} = 21.5$ min to the ephemeris uncertainty, so that it describes an individual future transit rather than the mean ephemeris. *TESS* is expected to observe TIC 402710280 in Sectors 113 and 114 (Burke et al. 2020); the transit of 2027 February 19 is predicted to fall in Sector 114. The transit of 2027 May 3 is observable in full from the southern hemisphere (e.g., La Silla Observatory). A transit absent from the predicted window would falsify the reported period.

9. SUMMARY

We report TIC 402710280.01, a $3.75\,R_\oplus$ sub-Neptune candidate with a 36.34-day period around the nearby K dwarf TIC 402710280, detected in four *TESS* full-frame-image transits spanning five years. The period is unique,

Table 4. Predicted transits of TIC 402710280.01

Epoch	T_c ($\mathrm{BJD_{TDB}}$)	σ_{pred} (min)
57	2461455.980	31
58	2461492.320	31
59	2461528.660	31
60	2461565.000	32
61	2461601.340	32
62	2461637.680	32
63	2461674.020	33
64	2461710.359	33
65	2461746.699	34
66	2461783.039	34

NOTE—Epochs count orbits from the Sector 39 transit. Epoch 57 (2027 February 19) is expected in *TESS* Sector 114; epoch 59 (2027 May 3, 03:43 UTC at La Silla) is fully observable at night from Chile.

the transit shape is consistent with the host star's density, the signal is located on the target or within 3.5″ of it, a blend with the only catalogued neighbour able to mimic it is strongly disfavoured, and the statistical false-positive probability is low. High-resolution imaging and radial velocities are needed to establish its planetary nature, and the transits predicted for 2027 provide a direct test of the ephemeris.

This paper includes data collected by the *TESS* mission, which are publicly available from the Mikulski Archive for Space Telescopes (MAST). Funding for the *TESS* mission is provided by NASA's Science Mission Directorate. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (https://www.cosmos.esa.int/gaia), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC, https://www.cosmos.esa.int/web/gaia/dpac/consortium). Funding for the DPAC has been provided by national institutions, in particular the institutions participating in the *Gaia* Multilateral Agreement. This research has made use of the Exoplanet Follow-up Observation Program (ExoFOP; DOI: 10.26134/ExoFOP5) website, which is operated by the California Institute of Technology, under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program, and of the NASA Exoplanet Archive, which is operated by the California Institute of Technology under the same program. Software development, data analysis and the drafting of this manuscript were assisted by an AI system (Claude, Anthropic); the author directed the work, verified the results and takes full responsibility for the content.

Facilities: TESS, Gaia

Software: astropy (Astropy Collaboration et al. 2022), astrocut (Brasseur et al. 2019), batman (Kreidberg 2015), emcee (Foreman-Mackey et al. 2013), lightkurve (Lightkurve Collaboration 2018), matplotlib (Hunter 2007), numpy (Harris et al. 2020), scipy (Virtanen et al. 2020), tess-point (Burke et al. 2020), TRICERATOPS (Giacalone & Dressing 2020)

DATA AVAILABILITY

The light curves of all transit windows, our photometry of Sectors 100–103, the measured transit times, the posterior summaries, and the analysis code are available on Zenodo (Vora 2026).

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